SPECTRALFORGE: A PHYSICS-BASED SIMULATOR FOR FLUORESCENCE HYPERSPECTRAL IMAGING

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ABSTRACT

Changes vs. accepted version: additions in red, deletions ~~struck through in grey~~. (Review copy; this file may exceed five pages. The submission file is `main.pdf`, 5 pages.) Summary: a new Sec. 2.6 and Fig. 6 compare the Eq. (1) model with real measurements: a drop at held-out excitations, and real collagen measured vs. predicted spectra at held-out excitations with a residual strip, with held-out error over 24 collagen cases (SAM $8.4 \pm 1.8^\circ$, shape $R^2 = 0.83 \pm 0.13$, relative RMS 0.097 ± 0.023; the empirical copy is slightly better, emission shape only). Also added: a two-level validity paragraph at the start of Sec. 3, a setup summary table (Table 1), a spectral-grid subsection (564 bands explained), the Studio workbench figure, and one abstract clause; Eq. (2) corrected (inner filter and reabsorption), several numbers replaced by values from the correct runs, previously uncited references cited, and the Evaluation, Benchmark and Discussion text tightened to stay at five pages (limitations content kept). Layout: figures enlarged (Figs. 1–6), captions in 9 pt. Acknowledgment: the United States R44 HL120511 award was removed.

Acquiring four-dimensional fluorescence hyperspectral data is slow and costly. An instrument records a full emission spectrum at each of several excitation wavelengths and for every pixel, fluorescence signals are weak and easily buried in scatter and autofluorescence, and the illumination needed to raise them photobleaches the sample, so a single cube can consume hours of scarce instrument time. This makes it hard to plan acquisitions, prototype processing methods, or build shared benchmarks. We present **SpectralForge**, an open, physics-based simulator that renders realistic 4D excitation-emission cubes with perfect ground truth from a parametric fluorophore model obeying the standard laws of fluorescence photophysics (Kasha, Vavilov, Franck-Condon, Stokes shift), with optional inner-filter attenuation, reabsorption, Foerster resonance energy transfer (FRET), scatter, photon and read noise, and instrument clutter. Because every rendered cube comes with its concentration maps and true emission peaks, a method can be developed, tested, and compared before any measurement is taken. As ~~one~~ a worked example, we ~~use the simulator to~~ study spectral band selection, where the goal is to keep only the informative bands and so acquire or process less. Scenes are built so that band variance is decoupled from class informativeness (measured correlation ≈ -0.1), so the task is not solved by variance ranking, and two controls, a random baseline and the full spectrum, tell a real gain from a chance one. ~~Running~~ Fitting the model to measured drop and collagen spectra tests it against real data, and running a panel of selectors through the simulator shows it behaving as designed and marks where label-free selection helps and where it does not. SpectralForge and its workbench are in active development, with early access ~~available to~~ for interested labs on request; the full study is released for reproducible use in 4D fluorescence imaging.

***Index Terms*—** fluorescence hyperspectral imaging, physics-based simulation, synthetic data, excitation-emission, photophysics, band selection.

1. INTRODUCTION

A multi-excitation fluorescence measurement illuminates a sample at several excitation wavelengths and, for each, records an emission spectrum~~across many bands~~. Every pixel therefore carries an excitation-emission matrix (EEM), and an image is a four-dimensional array over excitation, emission, height, and width (Fig. 1). ~~The modality is useful because different fluorophores~~Fluorophores, tissue states, ~~or~~ and species separate in this joint ~~excitation-emission~~ domain, which is why it is used ~~across~~ for biomedical and material imaging, for example label-free imaging of cardiac tissue by its endogenous autofluorescence [9,10]. ~~Adding excitation wavelengths turns each single spectrum into an excitation-emission matrix that separates overlapping fluorophores~~ Added excitations separate overlapping fluorophores that a single-excitation ~~hyperspectral~~ image cannot, ~~which improves downstream~~ improving tissue and material classification [12–14]~~. It is also expensive to acquire. Each added excitation~~, but each one multiplies the acquisition by a full emission scan, ~~fluorescence~~ signals are weak ~~and easily lost in~~ against scatter and autofluorescence, and ~~the illumination needed to raise them photobleaches the very molecules being measured~~illumination photobleaches the sample. A single ~~high-quality~~ 4D cube can ~~therefore~~ consume hours of ~~specialized instrument time~~, instrument time and labeled real cubes remain scarce.

~~This scarcity slows everything downstream.~~ Planning an acquisition, prototyping a processing method, or comparing algorithms all ideally happen before instrument time is committed, yet each needs representative data, and real cubes carry no ground truth ~~: there is no reference for~~ about which fluorophores are where or which bands ~~truly~~ matter. A physically realistic simulator ~~addresses both problems at once. If it~~ that renders cubes from the ~~actual~~ laws of fluorescence and returns the underlying concentrations and emission peaks ~~,~~ addresses both problems: a method can be built, tested, and benchmarked against an answer key, and only the promising design need ~~ever~~ reach the instrument.

This paper presents **SpectralForge**, an open simulator of that kind for 4D fluorescence imaging, with an interactive workbench for building scenes by hand and a small library for scripting them. To show the tool at work we take one concrete task, spectral band selection, and run a study through it end to end: the simulator provides the cubes and their ground truth, a controlled evaluation protocol scores the candidate selectors, and the results show both that the tool be-

haves as designed and where a learned selector actually helps. Band selection is only one use. The ; the same generator supports experiment planning, algorithm prototyping, teaching, and any study that benefits from 4D fluorescence data with a known answerprototyping, and teaching.

2. THE SPECTRALFORGE FORWARD MODEL

2.1. Photophysics

The generator is built on the physics of fluorescence [1, 3] so that "informative" means what it means physically. A fluorophore absorbs a photon, relaxes non-radiatively to the lowest vibrational level of the first excited singlet, and emits (the Jabłoński picture [5]). Four consequences shape the EEM. By *Kasha's rule* [2] emission occurs from that lowest level regardless of the excitation, so the emission shape is independent of excitation wavelength and the EEM is approximately trilinear (an excitation profile times an emission profile times a concentration map), the basis of PARAFAC [4]. By *Vavilov's rule* the quantum yield is, to first order, independent of excitation; only the amount absorbed changes. The *Franck-Condon* principle sets the broadened, roughly mirror-image band envelopes and an optional vibronic shoulder about $1300\,\mathrm{cm}^{-1}$ to the red. The *Stokes shift* red-shifts emission relative to absorption. Together these rules fix the structure of the forward model below; the departures from them that matter in practice (inner-filter attenuation, reabsorption, and FRET) enter as optional, switchable effects (Sec. 2.4), so that each one can be studied in isolation by turning it on or off.

2.2. Trilinear render and the additive invariant

Each fluorophore k is a parametric model with a peak-normalized Gaussian excitation profile $a_k(\lambda_{\mathrm{ex}})$ and an area-normalized emission profile $e_k(\lambda_{\mathrm{em}})$, extinction ε_k, quantum yield Φ_k, and a per-pixel concentration map $c_k(x, y)$. The rendered cube is a dilute linear superposition,

$$F(x, y, \lambda_{\mathrm{ex}}, \lambda_{\mathrm{em}}) = g \sum_k c_k(x, y)\, \varepsilon_k\, \Phi_k\, a_k(\lambda_{\mathrm{ex}})\, e_k(\lambda_{\mathrm{em}}), \quad (1)$$

where g folds lamp, exposure, and power. This is Kasha's rule written as algebra. The profiles are Gaussian , a_k peak-normalized and e_k area-normalized with widths $\mathrm{FWHM}/(2\sqrt{2\ln 2})$, and a fluorophore's brightness is the product Gaussian widths are $\sigma = \mathrm{FWHM}/(2\sqrt{2\ln 2})$, and brightness is $\varepsilon_k \Phi_k$. With all optional physics and artifacts off the model is *exactly linear*, $\mathrm{render}(A{+}B) = \mathrm{render}(A){+}\mathrm{render}(B)$, an invariant the implementation tests; difficulty is then added in realistic, opt-in increments. A per-pixel class label is the argmax over the discriminative fluorophores. Fig. 2 shows the fluorophore model.

2.3. Variance is decoupled from informativeness

The central design choice is the scene generator. Discriminative dyes are painted with smooth random concentration fields whose per-pixel argmax defines the class; they are dim and spectrally overlapping, so their content lives in low-variance band *shape*, while independent nuisance dyes are bright but class-irrelevant (Fig. 1c). Real photophysics compounds the decoupling: scatter ridgesand bright autofluorescenceScatter ridges, bright autofluorescence, and shot noise add high-variance but class-irrelevant signal, shot noise inflates bright redundant bands, and the discriminative signal sits in dim or subtly shaped bands. On a confounded scene the correlation between per-band variance and per-band discriminability is ≈ -0.1, versus strongly positive on naive clean data, so the highest-variance bands carry no class information are uninformative (Fig. 3) and a variance ranker cannot solve the taskvariance ranking fails.

2.4. Artifacts, nonlinear physics, and clutter

On top of the clean cube the generator adds Rayleigh scatter at $\lambda_{\mathrm{em}} = \lambda_{\mathrm{ex}}$ (with a second-order ghost at $2\lambda_{\mathrm{ex}}$) and a water Raman line at a fixed wavenumber offset ($\approx 3400\,\mathrm{cm}^{-1}$), each scaled by a per-pixel turbidity field, then Poisson shot noise and Gaussian read noise. Three optional effects introduce concentration nonlinearity: a primary inner filter $\exp(-sA_{\mathrm{ex}})\exp(-s_{\mathrm{ie}}A_{\mathrm{ex}})$; *reabsorption* $\exp(-sA_{\mathrm{em}}(\lambda))\exp(-s_{\mathrm{re}}A_{\mathrm{em}}(\lambda))$, whose overlap with the blue edge of emission reshapes the band and relocates discriminative information into band shape; and FRET, where a saturating fraction $E = kc_A/(1+kc_A)$ of donor energy is transferred to a co-localized acceptor, encoding dye co-localization in a band ratio. Compactly, the inner filter and transfer are

$$F = F_{\mathrm{obs}}\, 10^{-(A_{\mathrm{ex}}+A_{\mathrm{em}})/2}, \qquad E = \tfrac{k\,c_A}{1+k\,c_A},$$

attenuated signal and the transfer efficiency are

$$F_{\mathrm{obs}} = F\, e^{-s_{\mathrm{ie}}A_{\mathrm{ex}}}\, e^{-s_{\mathrm{re}}A_{\mathrm{em}}(\lambda_{\mathrm{em}})}, \qquad E = \frac{k\,c_A}{1+k\,c_A}, \quad (2)$$

with reabsorption $A_{\mathrm{em}}(\lambda) = \sum_k \varepsilon_k c_k\, a_k(\lambda)$ $A_{\mathrm{ex}} = \sum_k \varepsilon_k c_k a_k(\lambda_{\mathrm{ex}})$, dimensionless strength knobs $s_{\mathrm{ie}}, s_{\mathrm{re}}$ (not physical path lengths), reabsorption $A_{\mathrm{em}}(\lambda) = \sum_k \varepsilon_k c_k\, a_k(\lambda)$ (the excitation profile serves as absorption profile at the emission wavelength), and a Raman line at $1/\lambda_R = 1/\lambda_{\mathrm{ex}} - \Delta\bar{\nu}$. Finally, fixed-pattern clutter injects many class-irrelevant low-rank spatial fields on random band subsets, the analogue of illumination and detector artifacts and the dominant real-instrument confound. By dialing these knobs the generator it spans *clean* (variance $\approx$ informativeness, an idealized baseline that real acquisitions never meet), *realistic*, *clutter*, *nonlinear* (reabsorption or FRET), and *spatial-texture* regimes from a single in one call.

2.5. Ground truth, reproducibility, and real fluorophores

Every render returns a ground-truth sidecar with concentration maps, clean cubes, and per-fluorophore emission peaks, which enables a tight *peak-recovery* metric (did a selected band land within a tolerance of a true emission peak; a random selector scores about 0.33) in addition to broad information-retention metrics. The generator exposes a one-call study scripts (`build_dataset` API in `reports/`, not library API) call `render` with explicit knobs for size, emission sampling, nuisance brightness, scatter, noise, clutter, and nonlinear physics; a script re-runs every simulation and records exact parameters, so figures and numbers are reproducibleand record exact parameters. Parametric dyes can be replaced by real measured spectra imported from FPbase, and an interactive workbenchmeasured spectra from a downloaded FPbase payload via a library importer (not yet exposed in the workbench). The in-development 3D workbench, SpectralForge Studio (Fig. 4), lets a user paint concentration maps, render, and validate a selection place a lamp, camera, and samples in a 3D scene, render the 4D cube, and inspect it against ground truth. This makes SpectralForge usable both as a controlled benchmark and as a design tool for real 4D fluorescence experiments. As an interactive application (Fig. 4), a user picks or defines fluorophores in a library (Fig. **??**: excitation and emission peaks and widths,quantum yield, extinction), mixes them into materials that are concentration recipes, paints those materials onto a canvas as layers, then acquires, renders the 4D cube, selects bands, and validates against ground truth , all in onewindow. Each Studio scene is a plain JSON document with its seed and capture settings, so any capture can be saved, shared with collaborators and re-rendered exactly from that one file.

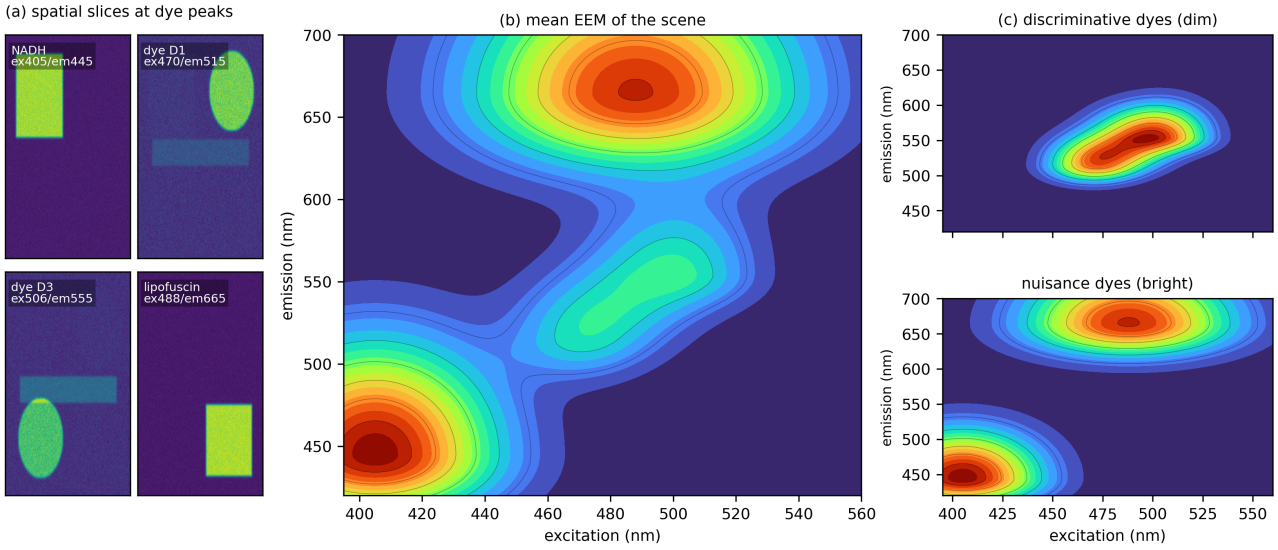


Fig. 1. A rendered 4D fluorescence cube from SpectralForge, ~~here~~ an example scene ~~painted in the workbench~~ with each dye laid down as a shape. (a) ~~A slice of the cube taken~~ Slices at each dye's emission peak ~~reveals that dye's~~ reveal its painted region~~(NADH square, dyes D1/D3 disks, lipofuscin block)~~; ~~a slice lights up only dyes that both absorb at its excitation and emit at its emission, so~~ every pixel effectively stores an ~~excitation-emission matrix (~~EEM~~)~~. (b) The mean EEM~~of the scene is the familiar fluorescence landscape~~: each fluorophore is a smooth blob at its excitation and (Stokes-shifted) emission. (c) ~~Decomposed by~~ By role, the discriminative dyes form ~~a single~~ one dim, overlapping cluster while the class-irrelevant nuisance dyes are bright and well separated, so band variance is dominated by ~~the~~ uninformative signal.

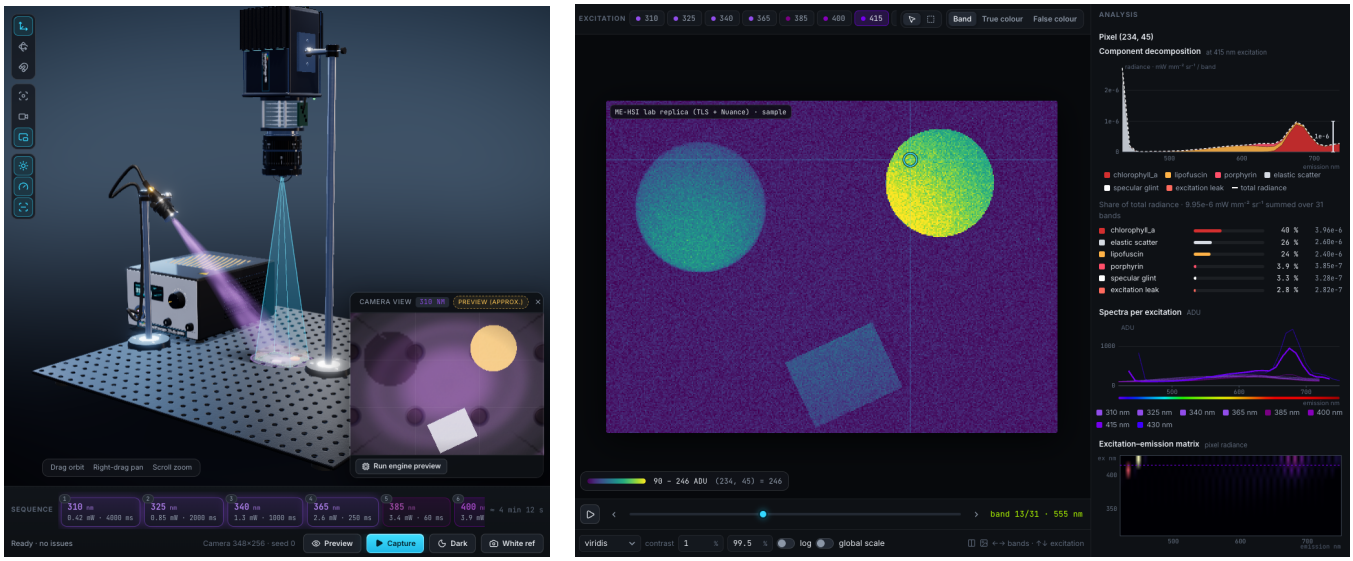


Fig. 4. SpectralForge Studio (in development). (a) Design: lamp, LCTF camera and samples placed in 3D, with the camera view. (b) Explore: a band image and, for the selected pixel, its ground-truth physical components, spectra and EEM. The study below used the core generator of Sec. 2, not Studio's added physics.

2.6. Validation against real acquisitions

Fitting Eq. (1), through a model of our tunable-lamp/LCTF instrument [12], to 4 excitations (310/340/385/415 nm) of 16 fluorescent drops predicts the 3 held-out excitations without gain refit (median $R^2 = 0.96$, SAM 6.0°, Fig. 6a). For collagen, whose intensities are incomparable across excitations, only shape is testable: after per-excitation gain and smooth throughput correction, held-out SAM is $8.4 \pm 1.8°$ and shape $R^2 = 0.83 \pm 0.13$ (24 cases; Fig. 6b,c), slightly worse than copying another sample's spectrum (7.7°).

3. EVALUATION PROTOCOL

Two questions are kept apart: simulator fidelity, covered by the linearity tests of Sec. 2.2 and the real-data check of Sec. 2.6; and benchmark validity, covered by known ground truth plus random and full-spectrum controls, which need no claim that a simulated cube equals a real one. Table 1 summarizes the setup.

3.1. Spectral grid

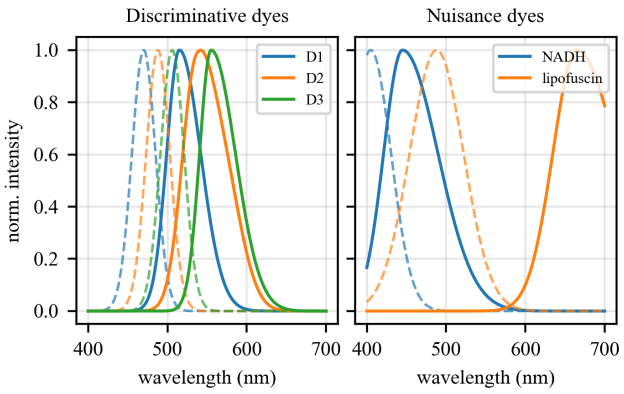


Fig. 2. Parametric fluorophore spectra ~~. Left: three discriminative dyes~~ (dashed, excitation; solid, emission)~~of modest brightness with~~ . Left: three dim, overlapping ~~emission and distinct excitation~~discriminative dyes. Right: two bright nuisance autofluorophores that dominate band variance but carry no class information.

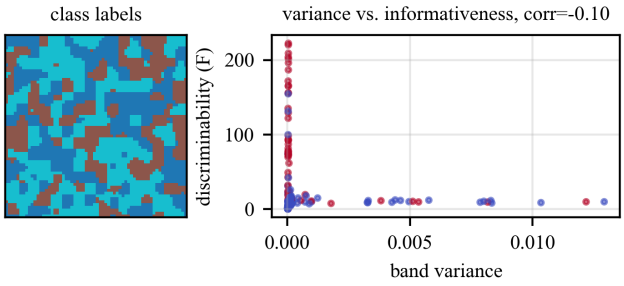


Fig. 3. Variance is decoupled from informativeness. ~~Each~~ Left: class labels. Right: each point is a band (red: ~~per-band variance against per-band discriminability. The~~ discriminative; blue: nuisance and scatter); the high-variance bands are nuisance ~~and~~ or scatter~~bands~~, not ~~the~~ discriminative ~~ones~~ (correlation near zero), so ~~ranking bands by~~ variance ranking fails.

The study uses 4 excitations (405, 470, 488, 506 nm) and emission from 420 to 700 nm: every 2 nm for Fig. 5 (141 bands, $4 \times 141 = 564$ excitation-emission bands, the "full-564" classifier) and every 5 nm for Table 2 (57 bands, 228 features).

Selection quality is the accuracy of a downstream classifier ~~trained~~ on the selected bands, ~~using labels~~ with labels used only at evaluation time. We score each subset with a classifier panel (k-nearest neighbors, random forest, MLP, logistic regression), cross-validated, reporting macro-F1 and, on nonlinear regimes, a *nonlinear-only gap* (the accuracy an expressive model recovers beyond a linear one), because a linear metric cannot credit a nonlinear-shape selection. Two controls matter here. A **random** baseline: a method that cannot beat the 95th percentile of random band subsets adds nothing. And the **full spectrum**: the accuracy ceiling a selector tries to approach. Where discrimination is spatial we use spatial-block cross-validation to remove spatial-autocorrelation leakage. ~~These two controls, often absent in prior work, guard against reporting a gain that a random subset would already reach.~~

Table 1. Experimental setup and the reason for each element.

Element	Setting	Why
Selectors	variance; PCA loadings; AE (label-free); mutual information (labels)	naive to supervised
Regimes	clean; realistic; clutter; reabsorption; FRET; spatial texture	ideal to nonlinear
Noise	5 levels: photon scale 5×10^4 to 200, Rayleigh 0.1 to 0.7, clutter 0 to 4.5	gain under degradation
Classifiers	k-NN, RF, MLP, LR; macro-F1; CV (block CV for texture)	scored by downstream use
Controls	random subsets (95th pct.); full spectrum	chance; ceiling
Seeds	6 (Table 2); 2 (Fig. 5); 3 (texture)	paired CIs
Scene, grid	64×64 px (Table 2: 40×40), 3 classes; 564 or 228 bands (Sec. 3.1)	known labels, peaks

4. USING THE BENCHMARK

Band selection, which removes non-informative bands to shrink a real hyperspectral acquisition ~~[11, 15]~~ [8, 11, 15] (its collagen and lichen data are public [16, 17]), is one thing SpectralForge is built to test, so we exercise it with a panel of standard selectors: variance ranking, PCA loadings [7], supervised mutual information (a labels-using reference), and a fully unsupervised ~~autoencoder-with-latent-perturbation~~ autoencoder [6]-with-latent-perturbation selector, which reconstructs the standardized cube, perturbs each latent dimension, and turns the per-band reconstruction sensitivity into a per-band influence. The ~~generator supplies the cubes and their ground truth; the protocol above scores each subset. The question is not which selector wins but whether the benchmark behaves as designed, that is, whether~~ question is whether decoupling variance from informativeness makes ~~it~~ the benchmark discriminate methods and regimes.

It does. On confounded scenes a variance ranker ~~barely beats random~~ does no better than, and usually below, random subsets (e.g. 0.40 vs 0.45 on the realistic regime). PCA loadings is the strongest blind selector only in the idealized *clean* regime, where the informative bands are also the brightest. No real fluorescence acquisition reaches that regime, since real cubes always carry scatter, autofluorescence, and noise; once those enter, on the realistic and FRET regimes, the two selectors tie. Label-free selection marginally overtakes PCA only where the physics places discriminative information in nonlinear band *shape*, as under reabsorption (macro-F1 0.449 vs 0.431, bootstrap 95% CI $[+0.001, +0.029]$, Cohen $d \approx +0.94$~~; its~~ , but paired Wilcoxon $p = 0.16$, so the gain is marginal; the AE-MLP subset carries nonlinear-only information ~~+0.030 that the linear selector's does not, −0.021~~ $+0.014$, the conv-AE -0.001, against -0.016 for PCA loadings; Table 2). The pattern is a general one: *informative implies has variance implies PCA-selectable*, so a nonlinear *classification* task need not imply a nonlinear *selection* task, and the benchmark separates the two. ~~This regime map,~~

Defining a fluorophore in the workbench library: choose a built-in dye or enter excitation and emission peaks and widths, quantum yield, and extinction. Materials are concentration mixtures of these fluorophores.

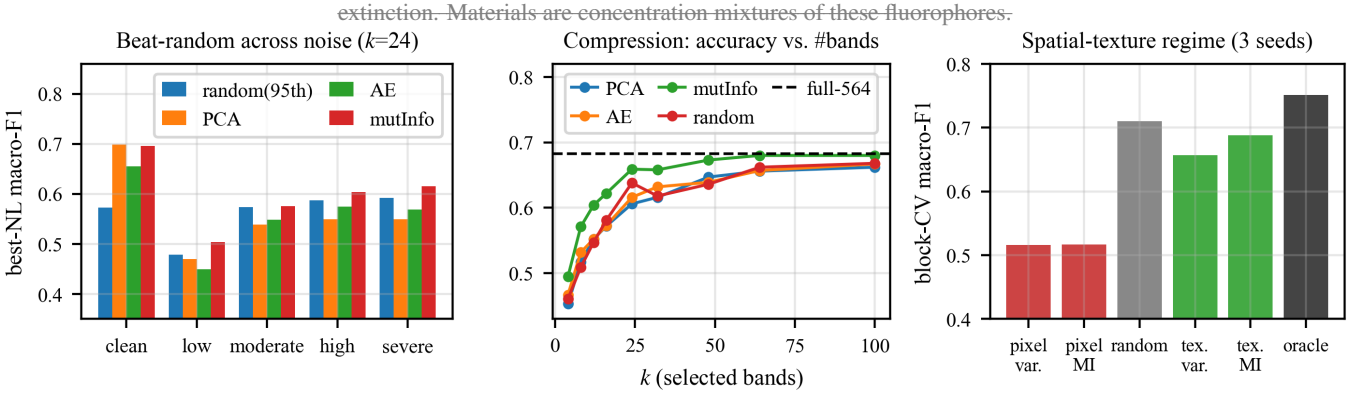


Fig. 5. The benchmark in use. Left: beat-random across five noise levels at $k = 24$ of 564 bands (Sec. 3.1): blind PCA and autoencoder beat the random 95th percentile only on clean data; under clutter only supervised MI does. Middle: compression over 4 to 100 bands; the autoencoder tracks PCA (about 95% of the full spectrum); labels compress about 2.7× better. Right: spatial texture (block CV): per-pixel selectors fall below random and no learned spatial selector beats it.

The SpectralForge workbench. Left: fluorophore recipes (materials) with a live spectrum. Centre: the painted canvas (a collagen and NADH patch, an FAD spot, an EGFP stripe), coloured by class label. Right: the layer stack. Bottom: acquire, render, and one-click validation against ground truth, here previewing a rendered emission band at 340 nm.

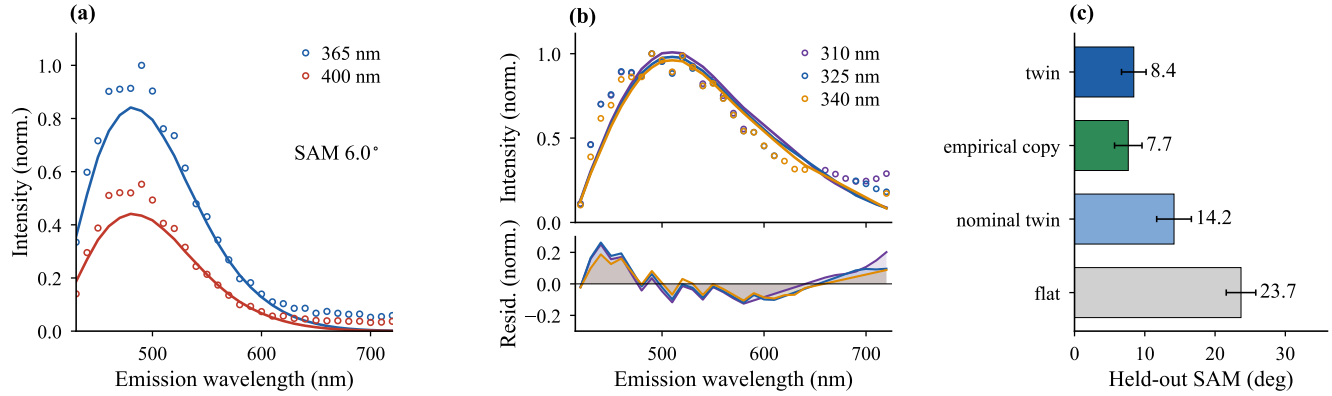


Fig. 6. Real-data validation. (a) Drop 5 (median SAM): measured (markers) and predicted (lines) at held-out excitations. (b) Collagen ROI 4 (median case of 24): measured, predicted and residual at held-out 310/325/340 nm. (c) Collagen held-out SAM, mean ± SD (24 cases).

Table 2. Label-free ~~autoencoder~~ AE vs. the strongest blind baseline (PCA loadings): best-of-panel macro-F1~~over~~ , six seeds~~. Bold marks the~~ ; bold is better~~selector~~. Clean margin: paired Wilcoxon $p = 0.03$; ~~the clean and~~ reabsorption~~margins are significant at~~ : bootstrap 95% ~~, the~~ CI excludes zero but $p = 0.16$ (marginal); rest are ties.

Regime	AE	PCA-load	verdict
clean	0.590	**0.635**	linear suffices
realistic	0.527	0.531	tie
reabsorption	**0.449**	0.431	~~label-free wins~~ marginal AE gain
FRET	0.635	0.642	tie

~~not a single winner, is what the benchmark reports.~~

~~The benchmark in use. Left: beat-random across five noise levels: blind PCA and autoencoder selection beat the random 95th percentile only on clean data, and under clutter only supervised mutual information does. Middle: compression curve: the autoencoder tracks PCA at every budget, both reaching about 95% of the full spectrum, while labels compress about 2.7× better. Right: the spatial-texture regime under block cross-validation: per-pixel selectors are the worst (below random) because the classes have identical per-pixel spectra, and only a spatial model with the right bands reaches the oracle.~~

The random and full-spectrum controls also mark the limits (Fig. 5). Under fixed-pattern clutter no blind selector beats a random subset, because clutter is high-variance but class-irrelevant; there only label-aware selection helps. No selector beats the full spectrum, so selection buys compression, not accuracy, and blind subsets are unstable band to band (Jaccard about 0.1 to 0.2), which favors selecting spectral regions or consensus sets. And when class is carried by spatial texture, so that every class shares the same per-pixel spectrum, per-pixel selection falls below random (three seeds, 0.52 ~~to 0.55,~~ against a random ~~0.674)while a spatial model with the right bands reaches the oracle (0.78). These are properties the benchmark makes visible, and they answer the practical questions: when to select bands, when to keep the spectrum, and when the problem is spatial rather than spectral~~0.71), the oracle bands reach 0.75, and none of the spatial selectors we tried (texture variance or MI, 0.66 to 0.69; a spatial CNN, 0.53 in a separate finer configuration with

random 0.58) beats random, so spatial selection remains open.

5. DISCUSSION AND CONCLUSION

SpectralForge gives 4D fluorescence imaging an open, physically realistic source of cubes with perfect ground truth, so ~~that~~ methods can be ~~designed, tested , and compared~~ tested before scarce instrument time is spent. ~~The band-selection study is one demonstration of that value. Paired with a protocol whose~~ With random and full-spectrum controls~~check that a reported gain is real~~, the simulator turns "does this selector help" into a measured, reproducible question, and its answer is specific: PCA loadings is the strongest blind baseline only in the idealized clean regime that no real acquisition reaches, ties label-free selection once realistic scatter and noise enter, and is ~~overtaken only~~ only marginally overtaken where the physics is nonlinear.

While SpectralForge captures key primary photophysical phenomena, the simulator operates under certain physical and optical simplifications. Currently, inner-filter attenuation and reabsorption are modeled using first-order exponential path approximations rather than full 3D radiative transfer or Monte Carlo photon transport through dense scattering media. Furthermore, the instrument model assumes an idealized optical train with spatially uniform illumination, uniform spectral detector response, and static photophysics, omitting dynamic exposure photobleaching kinetics during sequence acquisition as well as chromatic aberrations and thermal drift. Expanding the engine to incorporate full radiative transport and dynamic bleaching models remains an important direction for future extensions~~.~~; the in-development workbench (Fig. 4) already adds 3D Monte Carlo transport and photobleaching.

~~The same generator supports experiment planning, algorithm prototyping, and teaching.~~ A natural next step is to evaluate performance on complex, multi-component target samples (heterogeneous biological tissues, environmental specimens, or engineered multi-fluorophore composites), where informative spectral structure is decoupled from overall band variance. SpectralForge and its workbench are in active development toward a production release, and the authors can provide exclusive early access to interested laboratories and individuals on request (~~please~~ contact the corresponding author by email); the full experiment record ~~accompanying~~ for this study is released for reproducible use.

Acknowledgment

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